

Multi-App Hosting with AWS Application Load Balancer (ALB)

Theoretical Documentation

Introduction

This project demonstrates hosting multiple web applications on separate EC2 instances and accessing them through a single AWS Application Load Balancer using path-based routing.

Objective

Deploy Google and Drive web applications on separate EC2 instances and route requests using a single ALB.

AWS Services Used

Amazon EC2, Application Load Balancer (ALB), Target Groups, Security Groups, Apache HTTP Server.

How ALB Works

The Application Load Balancer operates at Layer 7 and routes traffic based on URL paths. Requests to /google are forwarded to the Google server and requests to /drive are forwarded to the Drive server.

Target Groups

Target Groups contain EC2 instances that receive traffic from the ALB. Separate target groups are created for each application.

Health Checks

ALB periodically checks instance health. If an instance becomes unhealthy, traffic is not forwarded to it.

Implementation Steps

1. Launch two EC2 instances. 2. Install Apache. 3. Deploy Google and Drive applications. 4. Create target groups. 5. Create ALB. 6. Configure path-based routing. 7. Test application access.

Path-Based Routing

/google → Google Target Group → Google EC2 Server
/drive → Drive Target Group → Drive EC2 Server

Advantages

Provides scalability, high availability, centralized traffic management, and support for multiple applications through a single endpoint.

Conclusion

This project demonstrates how AWS ALB can efficiently host and route traffic to multiple applications using path-based routing.